Light

GCSE Computer Science — Legal, Ethical & Environmental Issues

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

UK legislation requires organisations to protect personal data and gives individuals rights over how their information is used.
Which law provides these protections?

- ○ **A.** The Data Protection Act 2018 / UK GDPR
- ○ **B.** The Highway Code
- ○ **C.** The Electricity at Work Regulations
- ○ **D.** The Food Safety Act

Q2 (1 mark)

A person deliberately accesses a computer system without permission to view confidential files.
Which law has been broken?

- ○ **A.** The Copyright Act
- ○ **B.** The Computer Misuse Act 1990
- ○ **C.** The Freedom of Information Act
- ○ **D.** The Consumer Rights Act

Q3 (1 mark)

A student downloads a film from an illegal website without paying.
Which law protects the film creator's rights?

- ○ **A.** The Data Protection Act
- ○ **B.** The Health and Safety Act
- ○ **C.** The Copyright, Designs and Patents Act 1988
- ○ **D.** The Road Traffic Act

Q4 (1 mark)

A citizen requests to see documents about how their local council spent public money.
Which law gives them this right?

- ○ **A.** The Computer Misuse Act
- ○ **B.** The Copyright Act
- ○ **C.** The Data Protection Act
- ○ **D.** The Freedom of Information Act 2000

Q5 (1 mark)

Old computers and phones are discarded and end up in landfill sites where toxic materials leak into the soil.
What is this environmental problem called?

- ○ **A.** E-waste pollution
- ○ **B.** Sound pollution
- ○ **C.** Light pollution
- ○ **D.** Thermal pollution

Q6 (1 mark)

Large buildings containing thousands of servers that power websites and cloud services consume significant electricity.
What are these facilities called?

- ○ **A.** Shopping centres
- ○ **B.** Data centres
- ○ **C.** Sports stadiums
- ○ **D.** Car parks

Q7 (1 mark)

Some people cannot access online services because they lack internet connectivity, devices, or digital skills.
What is this inequality called?

- ○ **A.** The gender gap

- B. The generation gap
- C. The digital divide
- D. The wealth gap

Q8 (1 mark)

Software is released with its source code freely available, allowing anyone to view, modify, and distribute it.
What type of software is this?

- A. Proprietary software
- B. Shareware
- C. Trial software
- D. Open source software

Q9 (1 mark)

A website tracks users' browsing habits across the internet to build profiles for targeted advertising.
What ethical concern does this raise?

- A. Privacy concerns about collection and use of personal data without meaningful consent
- B. Noise pollution from computer fans
- C. Light pollution from screens
- D. Traffic congestion from delivery vehicles

Q10 (1 mark)

Robots and artificial intelligence systems increasingly perform tasks previously done by human workers.
What is a major ethical concern about this trend?

- ○ **A.** Robots consume too much water
- ○ **B.** Job displacement and unemployment as machines replace human workers
- ○ **C.** Robots are too colourful
- ○ **D.** Machines make too much noise

Q11 (1 mark)

Streaming a video requires data to travel through networks and be processed by servers, all of which use electricity.
What environmental impact does this create?

- ○ **A.** Reduced water levels in rivers
- ○ **B.** Increased soil erosion
- ○ **C.** Carbon emissions from the electricity generation powering digital infrastructure
- ○ **D.** Reduced biodiversity in forests

Q12 (1 mark)

A photographer shares their images online with a licence allowing others to use them freely as long as they give credit.
What type of licence is this likely to be?

- ○ **A.** A top-secret government licence

- **B.** A driving licence
- **C.** A fishing licence
- **D.** A Creative Commons licence

Q13 (2 marks)

A company collects customer email addresses for order confirmations. They then sell this data to marketing companies without telling customers.
Which data protection principle has been violated?

- **A.** Data should only be used for the specific purpose for which it was collected; using it for marketing without consent violates purpose limitation
- **B.** Data must be stored in alphabetical order
- **C.** Data must be printed on recycled paper
- **D.** Data must be stored for at least 100 years

Q14 (2 marks)

An employee uses their legitimate work login to access files they are not authorised to view, looking at colleagues' salary information.
Has a Computer Misuse Act offence been committed?

- **A.** No, because they used valid login credentials
- **B.** Yes — accessing data beyond one's authorisation level is unauthorised access even with valid credentials; authorisation is about what you can access, not just authentication
- **C.** No, because salary information is public

- ○ **D.** No, because no damage was caused

Q15 (2 marks)

An employer installs software on company computers that captures screenshots every 5 minutes and logs all keystrokes.
What ethical issues does this raise?

- ○ **A.** The software might slow down the computers slightly
- ○ **B.** The screenshots use too much storage space
- ○ **C.** The software is difficult to install
- ○ **D.** Intensive surveillance damages trust, may capture personal information (banking, health), creates psychological stress, and treats employees as suspects rather than trusted team members

Q16 (2 marks)

A technology company designs phones with batteries that cannot be replaced, encouraging customers to buy new devices when batteries degrade.
What environmental concerns does this practice raise?

- ○ **A.** The phones are too thin to hold comfortably
- ○ **B.** The phones are too light to feel premium
- ○ **C.** Planned obsolescence shortens product lifespan, increases e-waste, wastes resources used in manufacturing, and contributes to environmental damage from mining rare materials
- ○ **D.** The phones are too quiet during calls

Q17 (2 marks)

An AI system used for recruitment is found to rate female candidates lower than equally qualified male candidates.
What does this demonstrate about AI systems?

- ○ **A.** AI systems can perpetuate and amplify biases present in their training data; if historical hiring data favoured men, the AI learns and reproduces this discrimination
- ○ **B.** AI systems are always completely fair and objective
- ○ **C.** Female candidates should not apply for technology jobs
- ○ **D.** AI cannot be used for recruitment purposes

Q18 (2 marks)

A student uses 30 seconds of a copyrighted song in their educational video project posted to YouTube.
Does this infringe copyright?

- ○ **A.** No, because it is for educational purposes only
- ○ **B.** No, because 30 seconds is too short to count
- ○ **C.** Using any portion of copyrighted work without permission may infringe copyright; 'fair dealing' exceptions for education are narrow and posting publicly on YouTube likely exceeds them
- ○ **D.** No, because YouTube allows all copyrighted content

Q19 (2 marks)

A company wants to reduce the environmental impact of its IT operations.
Which approach would be most effective?

- ○ **A.** Print all documents to reduce screen time
- ○ **B.** Replace all equipment annually with newer models
- ○ **C.** Use larger monitors to reduce eye strain
- ○ **D.** Extend device lifespan, use renewable energy, implement virtualisation to reduce servers, and recycle equipment properly

Q20 (2 marks)

A website is designed without consideration for users with visual impairments — images have no descriptions, and the layout does not work with screen readers.
What ethical principle has been neglected?

- ○ **A.** The website should have more advertisements
- ○ **B.** Accessibility — designing technology to be usable by people with disabilities is both an ethical obligation and often a legal requirement
- ○ **C.** The website should load faster
- ○ **D.** The website needs more animations

Q21 (2 marks)

A school chooses between proprietary office software (with annual licence fees) and open source alternatives (free to use).
What factors should inform this decision?

- ○ **A.** Only the colour of the software interface matters
- ○ **B.** Decisions should only consider which software has more fonts

○ **C.** Total cost (including support and training), compatibility with existing systems, available features, support options, and long-term sustainability of each option

○ **D.** Only the number of keyboard shortcuts matters

Q22 (2 marks)

Social media algorithms prioritise content that generates strong emotional reactions, which often includes misleading or divisive content. What ethical concerns does this raise?

○ **A.** Social media platforms use too much purple in their design

○ **B.** Social media apps take up too much phone storage

○ **C.** Social media notification sounds are too quiet

○ **D.** Engagement-driven algorithms can spread misinformation, polarise public discourse, affect mental health, and prioritise profit over societal wellbeing

Q23 (2 marks)

A UK website collects personal data from users in the USA, Europe, and Asia.
What legal complexity does this create?

○ **A.** Time zones make website maintenance difficult

○ **B.** Different countries have different data protection laws; the website must comply with regulations in each jurisdiction where users are located, which may have conflicting requirements

○ **C.** Different countries use different keyboard layouts

○ **D.** Different countries have different electricity voltages

Q24 (3 marks)

A hospital uses an AI system to prioritise patients for treatment. The system considers factors including age, pre-existing conditions, and predicted recovery probability.
What ethical issues must be considered when deploying such a system?

○ **A.** Only the speed of the AI system matters for healthcare

○ **B.** AI decisions about healthcare are always superior to human judgment

○ **C.** Healthcare AI is too expensive to evaluate ethically

○ **D.** Accountability (who is responsible for AI decisions?), transparency (can decisions be explained?), bias (does the AI disadvantage certain groups?), human oversight (should humans review AI recommendations?), and the fundamental question of whether life-affecting decisions should be delegated to algorithms

Q25 (3 marks)

A company claims its new smartphone is 'eco-friendly' because:
• The packaging is made from recycled cardboard
• The charger uses slightly less energy
• The phone can be traded in for recycling
However, the phone has a non-replaceable battery, is difficult to repair, and will likely be replaced within two years.
What does this reveal about assessing environmental claims?

○ **A.** Full lifecycle assessment is needed — manufacturing impact, lifespan, repairability, and end-of-life processing matter more than minor packaging improvements; focusing on small positives while ignoring major negatives is 'greenwashing'

○ **B.** Packaging is the only environmental consideration that matters

○ **C.** Battery design has no environmental impact

○ **D.** Shorter product lifespan is better for the environment

Q26 (3 marks)

After a terrorist attack, the government proposes legislation requiring technology companies to provide 'backdoor' access to encrypted communications.
How should the competing interests be evaluated?

○ **A.** Security services should have unlimited access to all communications without any restrictions

○ **B.** Encryption should be banned entirely to prevent criminal use

○ **C.** Law enforcement's need to investigate serious crimes must be balanced against privacy rights, security risks of backdoors (which criminals could also exploit), effects on journalism and activism, economic impact on technology industry, and whether such measures would actually be effective given globally available encryption

○ **D.** Technology companies should make all decisions about surveillance without government input

Q27 (3 marks)

Self-driving vehicles must be programmed to respond to unavoidable accident scenarios.
For example: if a crash is inevitable, should the car prioritise protecting its passengers or minimising total harm (which might mean sacrificing the passenger to save more pedestrians)?
What ethical challenges does this present?

○ **A.** Autonomous vehicles have no ethical implications because computers make decisions

○ **B.** Programming life-or-death decisions into algorithms raises questions about moral responsibility, whose values are encoded, transparency of decision-making, legal liability, and whether such trolley-problem scenarios are realistic or distract from more common safety decisions

○ **C.** Ethical considerations should never influence vehicle programming

○ **D.** Manufacturers should never be held accountable for autonomous vehicle decisions

Q28 (3 marks)

A social media company headquartered in the USA operates globally. UK regulators want to enforce UK laws against harmful content, but the company argues it only needs to follow US law.
What challenges does this present for regulating technology companies?

○ **A.** Regulation is unnecessary because technology companies self-regulate effectively

○ **B.** All countries should simply adopt US laws to avoid conflicts

○ **C.** Technology operates within single jurisdictions with clear legal boundaries

○ **D.** Global technology platforms create jurisdictional challenges: companies may structure operations to minimise regulatory exposure, enforcement across borders is difficult, different countries have conflicting legal standards, and platforms may threaten withdrawal from markets rather than comply

Q29 (3 marks)

Emerging technologies like brain-computer interfaces could eventually allow direct thought communication, memory enhancement, or emotional state modification.

What ethical frameworks should guide development of such technologies?

- ○ **A.** Any technology that is possible should be developed without ethical constraints
- ○ **B.** Emerging technologies should be banned entirely to avoid ethical questions
- ○ **C.** Proactive ethical frameworks should address consent (especially for mental autonomy), equality of access (avoiding cognitive divides), identity and authenticity (what makes us 'us'), potential for coercion or control, and long-term societal effects — considered before widespread deployment rather than after
- ○ **D.** Ethics are irrelevant to technology development

Q30 (3 marks)

The technology industry's environmental impact includes:
• Resource extraction (rare earth mining, water usage)
• Manufacturing (energy-intensive, global supply chains)
• Operation (data centres, network infrastructure)
• Disposal (e-waste, hazardous materials)
Who should bear responsibility for addressing these impacts?

- ○ **A.** Only individual consumers should be responsible for environmental choices
- ○ **B.** Responsibility should be shared: manufacturers should design sustainable products, governments should regulate and incentivise, consumers should make informed choices, and international cooperation should address global supply chains — no single actor can address systemic issues alone
- ○ **C.** Environmental impacts are too complex to address
- ○ **D.** Only governments should be responsible with no role for industry or consumers

Total: **55** marks • Time: **30 min**

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